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Session Date: May 12, 2026 (1 hour)

Objective: Run the pipeline at N=1 and N=4 then observe the difference in timing.

Experiments/Analysis:

Running the same trace at both thread counts gave the following results:

N=1: 1010.87 ms

N=4: 1080.47 ms

Results showed that N=4 was slower. While this was unexpected, further thought showed that this was actually the expected result. The trace has 5 D entries that have a combined sleep time of ~1000ms. The pace was set by the producer and not by the consumers. After each burst of pushes, the consumers drained the queue and became idle, awaiting the next burst. At N=1, the one consumer was more than enough to manage the queue. The contention caused by N=4, where 4 consumers were all calling pop on a mutex-protected queue, resulted in a slower system, due to a significant amount of contention and no real benefit of throughput.

Solutions/Workarounds:

No changes were needed and both runs were the same as the expected weight sum of 834418737.

Learning Outcome:

The pace of the system in this case was set by the producer. Because the producer was largely sleeping, the consumers did not need to be active. In cases like this, performance is improved when the number of threads is reduced to the point where each thread is doing enough work to justify the overhead of the thread rather than competing for a synchronized resource, such as a queue. Performance could also be improved by changing the queue so that instead of one item each push or pop, a batch of several items is processed.